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Project Title: Banking Database Management System using SQL

□ INTRODUCTION

This project is a Banking Database Management System designed using SQL.

It simulates real-world banking operations such as customer management, account handling, loan processing, and transaction tracking.

The objective of this project is to demonstrate relational database design and SQL query skills for data analysis.

□ OBJECTIVE

The main objective of this project is to:

- Manage banking data efficiently
- Maintain relationships between customers, accounts, loans, and transactions
- Perform data analysis using SQL queries
- Extract meaningful insights from financial data

□ DATABASE TABLES

The project consists of 4 main tables:

1. Customers Table:

- customer_id (Primary Key)
- name
- age
- city

2. Accounts Table:

- account_id (Primary Key)
- customer_id (Foreign Key)

- account_type
- balance

3. Loans Table:

- loan_id (Primary Key)
- customer_id (Foreign Key)
- loan_amount
- status

4. Transactions Table:

- transaction_id (Primary Key)
- account_id (Foreign Key)
- amount
- transaction_type
- transaction_date

□ SQL QUERIES SUMMARY

The project includes 36 SQL queries:

- Basic Queries (1–8): Data retrieval
- Intermediate Queries (9–17): Aggregation and grouping
- Advanced Queries (18–30): Business analysis
- Join Queries (31–34): Table relationships
- Views (35–36): Data abstraction and summaries

□ KEY FEATURES

- Customer management system

- Account tracking system
- Loan management system
- Transaction monitoring system
- Data analysis using SQL

INSIGHTS / FINDINGS

- Some customers have significantly higher balances than others
- Certain cities show higher banking activity
- Some accounts are inactive with no debit transactions
- Some customers are high-risk due to low balance and loans
- Credit transactions are more frequent in some accounts

CONCLUSION

This project demonstrates the working of a banking system using SQL.

It successfully manages banking data and provides meaningful insights.

It helps in understanding real-world database operations such as relationships, queries, and data analysis.

FINAL NOTE

This project helped in understanding SQL concepts like joins, grouping, aggregation, and views.

It also improved problem-solving skills in database management systems.